

FUSHIKI, Japan

WMO: 476060

Lat: 36.7922N

Lon: 137.0561E

Elev: 13

StdP: 101.17

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) -1.5	(d) -6.5	(e) 2.2	(f) 2.4	(g) -5.4	(h) 2.4	(i) 2.0	(j) 8.7	(k) 4.4	(l) 7.6	(m) 4.2	(n) 2.5	(o) 230	(p) 0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.7	(c) 33.6	(d) 25.1	(e) 32.1	(f) 25.1	(g) 30.6	(h) 24.8	(i) 26.6	(j) 30.7	(k) 26.0	(l) 30.0	(m) 25.4	(n) 29.3	(o) 3.8	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.7	28.8	24.8	19.9	28.4	24.2	19.2	27.9	83.2	30.8	80.5	30.0	78.0	29.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.4	5.5	-4.1	36.5	1.1	1.2	-4.9	37.3	-5.6	38.0	-6.2	38.7	-7.1	39.5
			-4.8	27.2	1.0	0.8	-5.5	27.7	-6.1	28.2	-6.7	28.6	-7.5	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	2.8	3.3	6.7	11.9	17.2	21.0	25.6	27.2	22.9	17.3	11.2	5.7
	DBStd	8.80	2.48	2.80	3.32	3.49	2.91	2.37	2.71	2.27	2.79	2.89	3.23	3.15
	HDD10.0	702	222	188	112	19	0	0	0	0	0	0	24	136
	HDD18.3	2177	480	421	359	196	58	4	0	0	1	54	215	390
	CDD10.0	2334	0	1	11	75	222	331	484	533	388	226	59	4
	CDD18.3	768	0	0	0	2	21	84	226	275	139	21	0	0
	CDH23.3	5903	0	0	1	22	126	297	1830	2784	785	58	1	0
	CDH26.7	1902	0	0	0	4	21	58	614	1014	182	9	0	0
(14) Wind	WSAvg	2.6	2.6	2.7	2.9	2.9	2.7	2.5	2.4	2.4	2.6	2.6	2.6	2.7
(15) Precipitation	PrecAvg	2262	261	164	149	125	123	165	215	182	204	161	229	284
	PrecMax	2721	445	281	217	230	308	327	526	472	381	278	449	609
	PrecMin	1581	121	54	70	38	46	53	40	41	79	88	146	163
	PrecStd	277	72	55	42	52	66	65	130	105	74	45	62	102
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	16.0	21.4	26.3	29.0	30.9	35.3	35.6	33.4	27.7	22.2	16.7
		MCWB	7.9	10.7	11.9	15.0	18.3	22.7	25.0	25.2	24.1	20.2	15.2	11.5
	2%	DB	9.7	11.9	17.6	22.6	26.6	28.3	33.4	33.9	30.5	24.7	19.7	13.9
		MCWB	6.6	7.7	10.4	14.2	17.1	21.0	25.0	25.3	24.1	19.1	14.7	10.5
	5%	DB	8.2	9.6	14.6	20.0	24.4	26.4	31.7	32.6	28.7	23.1	17.9	12.1
		MCWB	5.6	6.5	9.3	12.9	16.4	20.8	24.8	25.4	23.5	18.4	13.8	9.2
	10%	DB	6.9	8.0	12.2	17.6	22.5	24.8	30.0	31.3	27.1	21.8	16.3	10.6
		MCWB	4.8	5.2	8.4	12.2	16.2	20.5	24.6	25.4	22.8	17.6	13.1	8.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	11.8	13.4	17.2	20.6	24.1	26.8	27.6	26.1	22.1	17.1	13.2
		MCDB	11.1	14.4	18.4	22.2	25.2	28.0	31.5	31.5	29.7	25.4	18.9	14.7
	2%	WB	7.1	8.6	11.5	15.4	19.0	23.0	26.1	27.0	25.0	20.7	15.8	11.0
		MCDB	9.1	11.2	15.8	20.3	23.2	26.0	30.5	31.0	28.7	23.3	18.7	13.4
	5%	WB	6.0	6.9	10.0	14.1	17.9	22.0	25.5	26.4	24.1	19.5	14.5	9.5
		MCDB	7.8	9.1	13.4	18.2	22.0	25.0	29.7	30.4	27.5	21.9	17.2	11.7
	10%	WB	4.9	5.4	8.7	12.9	17.0	21.1	25.0	25.8	23.4	18.3	13.3	8.2
		MCDB	6.8	7.7	11.8	16.5	20.9	23.9	29.0	29.8	26.4	21.0	16.0	10.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	6.2	8.0	8.5	8.1	6.1	6.1	6.7	6.3	7.2	7.1	5.9
		MCDBR	7.8	9.9	12.8	13.3	12.2	8.9	8.9	8.9	8.3	8.5	9.7	8.5
	5% WB	MCWBR	5.5	6.8	7.5	6.8	5.5	4.0	3.3	3.4	3.6	4.3	5.8	5.9
		MCWBR	7.3	9.1	10.8	10.1	9.1	7.0	7.6	7.4	7.0	6.8	8.1	7.9
(40) Clear-Sky Solar Irradiance	taub		0.420	0.449	0.511	0.536	0.538	0.558	0.549	0.530	0.478	0.444	0.425	0.421
		taud	1.968	1.914	1.838	1.858	1.920	1.951	2.039	2.098	2.198	2.182	2.136	2.023
	Ebn at Noon		721	757	750	759	766	749	753	757	776	761	721	691
		Edh at Noon	137	164	194	200	191	186	169	156	134	124	116	123
(44) All-Sky Solar Radiation	RadAvg		1.61	2.43	3.55	4.60	5.26	5.01	4.78	4.96	3.88	3.11	2.04	1.41
	RadStd		0.16	0.21	0.28	0.38	0.57	0.38	0.71	0.53	0.30	0.26	0.18	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	+0.60	+0.90	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page